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RESEARCH ARTICLE

IP SafeGuard—An AI-Driven Malicious IP Detection Framework

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ABSTRACT The rising frequency and sophistication of cyberattacks have made real-time malicious IP detection a critical challenge for modern Security Operations Center (SOC). Traditional solutions, such as static blacklists and manual IP reputation checks, are no longer sufficient in today's dynamic threat scenario. To overcome these constraints, we present **IP SafeGuard**, an AI-driven platform that incorporates multi-source threat intelligence, sophisticated feature engineering, and machine learning (ML) for real-time IP categorization. The framework collects data from **AbuseIPDB**, **VirusTotal**, and other sources to compute a Dynamic Threat Score (DTS) for each IP address. It leverages an XGBoost-based classification model to achieve high accuracy and low false-positive rates, even in skewed datasets. Experimental findings indicate the improved performance of **IP SafeGuard**, with an accuracy of 98.2%, a precision of 97.8%, and a recall of 98.5%. The average detection duration of 45 milliseconds makes it appropriate for real-time SOC integration, enabling automated incident response through Security Information and Event Management (SIEM) alerting and firewall blocking. The framework's modular design assures scalability and adaptability, making it a vital tool for high-volume situations. By overcoming the limits of old approaches and using the power of ML, **IP SafeGuard** considerably boosts the efficiency and efficacy of current cybersecurity systems. Future work involves expanding the system to enable new threat intelligence sources and studying federated learning for secure and privacy-preserving threat information exchange.

INDEX TERMS Cyber security, cyber attack, IP address, artificial intelligence, threat intelligence, SOC, cyber defense.

I. INTRODUCTION

In the exponential increase of cyber threats, including Distributed Denial-of-Service (DDoS) assaults, botnets, phishing campaigns, and zero-day vulnerabilities, has made real-time threat identification a major issue for current cybersecurity systems [1]. SOC is responsible for monitoring and protecting networks against these attacks, but they typically struggle with the sheer number and complexity of

hostile activity [2]. Among these problems, the discovery and categorization of malicious IP addresses remain a cornerstone of effective cybersecurity protection. Conventional methods, such as static blacklists and manual IP reputation assessments, are inadequate in the current volatile threat environment, where adversaries often alter strategies and exploit emerging weaknesses. [3], [4].

Static blacklists, however advantageous for prohibiting recognized malicious IPs, have some limitations. They need frequent updates, have limited real-time adaptability, and often produce high false-positive rates, which might

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disrupt genuine traffic [5]. Manual IP reputation assessments are laborious and ineffective, particularly in fast-paced environments such as SOC. These constraints underscore the necessity for an automated, scalable, and accurate solution for malicious IP detection [6].

Recent improvements in online threat intelligence platforms, such as AbuseIPDB and VirusTotal, have provided valuable resources for IP reputation analysis. Nonetheless, these platforms are often used in isolation, resulting in disjointed and incomplete danger evaluations [7]. Moreover, the integration of ML techniques into cybersecurity has showed promise, although existing ML-based solutions frequently struggle with high computing costs, unbalanced datasets, and limited real-time applicability [8]. These shortcomings in the existing state-of-the-art underline the need for a comprehensive framework that integrates multi-source threat information with powerful ML for real-time malicious IP identification.

In this study, we introduce **IP SafeGuard**, an innovative AI-driven platform aimed to overcome these difficulties. **IP SafeGuard** combines threat intelligence from numerous sources, including AbuseIPDB,¹ VirusTotal,² Cisco Talos,³ and internal network logs, to give a comprehensive picture of IP reputation. The system employs a **XGBoost** ML model for precise and effective classification, and it makes use of sophisticated feature engineering techniques to extract pertinent insights from raw data. Furthermore, **IP SafeGuard** is designed to integrate seamlessly with SOC operations, allowing for real-time threat mitigation and automated incident response.

The key contributions of this research are as follows:

- 1) **Multi-Source Adaptive Threat Intelligence Fusion:** The article presents a comprehensive methodology for synthesizing data from several threat intelligence sources to calculate a DTS for each IP address.
- 2) **Advanced Feature Engineering:** A thorough feature extraction technique is shown to convert unrefined threat intelligence data into organized features appropriate for machine learning models.
- 3) **ML-Based Classification:** The suggested XGBoost-based classifier attains elevated accuracy and minimal false-positive rates in real-time threat detection contexts.
- 4) **SOC Integration and Automation:** The framework is engineered to integrate with current SOC technologies, such as SIEM systems and firewalls, facilitating automated threat detection and response.

The remainder of this paper is organized as follows: Section II reviews related work in IP threat detection and ML-based cybersecurity. Section III details the methodology and design of **IP SafeGuard**. Section IV presents the

experimental results and discusses their implications. Finally, Section V concludes the paper and outlines future research directions.

II. RELATED WORK

The detection and classification of malicious IP addresses have been extensively studied in the field of cybersecurity. This section reviews existing research on IP threat detection, ML-based approaches, and threat intelligence integration, highlighting the gaps that **IP SafeGuard** aims to address.

A. TRADITIONAL IP THREAT DETECTION METHODS

Traditional approaches for IP threat identification depend mainly on static blacklists and rule-based systems. D Likhomanov et al. presented a static blacklisting solution employing predetermined lists of dangerous IPs. While successful for recognized dangers, this strategy lacks the flexibility to new or changing threats [9]. Similarly, A Hussain et al. examined the use of firewalls and intrusion detection systems (IDS) to prevent malicious traffic [10]. However, these systems generally suffer from high false-positive rates and require frequent upgrades to stay successful. Salah and Chaudary established a rule-based system for IP filtering, however their method suffered with scalability in big networks [11]. Chen et al. emphasized the inefficiencies of manual IP reputation checks, underlining the necessity for automation in current SOC [12]. Davis et al. developed feature engineering approaches for IP threat detection, however their approach was constrained by the absence of real-time capabilities [13], [14].

B. ML-BASED APPROACHES

Machine learning has developed as an effective technique for identifying bogus IP addresses in dynamic environments. Assiri developed a random forest-based model for IP classification, achieving commendable accuracy but using considerable computing resources [15]. Dong deployed support vector machines (SVM) to classify IPs based on network traffic characteristics, demonstrating increased accuracy but constrained scalability [16].

More recently, Tuan et al. applied deep learning methods to detect malicious IPs, obtaining state-of-the-art performance but at the expense of increased complexity and training time [17]. Leevy et al. explored the use of XGBoost for IP classification, confirming its effectiveness in handling imbalanced datasets [18]. Bayazit et al. offered a neural network-based approach for IP threat detection, although their model required large training data [19]. Umer and Sher applied clustering algorithms to identify malicious IPs, yet their approach struggled with high-dimensional data [20]. Nair et al. suggested an anomaly detection framework for IP threat detection, although their system was limited by large false-positive rates [21].

¹<https://www.abuseipdb.com>

²<https://www.virustotal.com>

³<https://talosintelligence.com>

C. THREAT INTELLIGENCE INTEGRATION

The integration of threat intelligence from numerous sources has been found to boost the accuracy of IP threat detection. Lewis et al. proposed a methodology that integrates data from **AbuseIPDB** and **VirusTotal** to enhance IP reputation scoring [7]. However, their technique lacked real-time capabilities and was restricted to offline analysis. Sun et al. created a multi-source threat intelligence platform that aggregated data from numerous feeds, however their system battled with data inconsistency and excessive latency [22]. Aminu et al. proposed a dynamic threat intelligence framework, however their solution was not optimized for real-time environments [23]. Ranjan and Dahiya studied the use of APIs for threat intelligence integration, however their method was hampered by API rate constraints and data quality difficulties [24]. Chatziamanetoglou and Rantos presented a blockchain-based threat intelligence sharing platform, however their system needed substantial computing resources [25]. Sarhan et al. presented federated learning for threat intelligence sharing, however their technique was constrained by communication overhead [26]. Rehman and Hashmi offered a real-time threat intelligence architecture, however their system suffered with scalability [27].

D. REAL-TIME DETECTION AND SOC INTEGRATION

Real-time detection and integration with SOC are important for effective cybersecurity. Singh et al. constructed a real-time IP categorization system exploiting streaming data, however their solution was confined to small-scale networks [28]. Gschwandtner et al. studied the integration of threat intelligence with SOC tools, such as SIEM systems, while their architecture lacked automated reaction capabilities [29]. Manda presented an automated incident response system, although their technique was not suitable for high-volume scenarios [30]. Morariu et al. provided a scalable framework for real-time IP detection, while their system needed considerable computer resources [31]. Farooq and Otaibi presented an efficient algorithm for real-time threat detection, yet their technique was hampered by its dependence on a single threat intelligence source [32]. Saeed suggested an adaptive architecture for real-time IP detection, yet their system struggled with shifting threat landscapes [33]. Uccello et al. offered a hybrid strategy combining rule-based and ML-based techniques, although their system required extensive tuning [34].

E. GAPS IN EXISTING RESEARCH

Despite these advancements, several gaps remain in the current state-of-the-art:

- **Limited Real-Time Capabilities:** Many present systems are built for offline analysis and cannot handle real-time data streams.
- **High computing Costs:** ML methods, especially deep learning, can need large computing resources, restricting their application in resource-constrained contexts.

- **Incomplete Threat information:** Most systems rely on a single or restricted number of threat information sources, resulting in incomplete or fragmented threat assessments.
- **Lack of SOC Integration:** Few solutions are intended for smooth integration with SOC procedures, limiting their practical usefulness.

F. CONTRIBUTION OF IP SAFEGUARD

IP SafeGuard addresses these gaps by integrating multi-source threat information, innovative feature engineering, and ML for real-time malicious IP identification. Unlike prior systems, **IP SafeGuard** is intended for easy connection with SOC tools, enabling automated incident response and real-time threat mitigation. The framework's modular architecture enables scalability and versatility, making it appropriate for high-volume situations.

III. METHODOLOGY

The **IP SafeGuard** framework is built on four basic algorithms that work synergistically to identify and categorize dangerous IP addresses in real-time as shown in Fig. 1. These algorithms are meant to automate the process of Adaptive Threat Intelligence Fusion, feature extraction, ML-based categorization, and SOC integration, assuring a robust and scalable solution for contemporary SOC. Below, we present each technique in depth, complete with pseudocode, flowcharts, and further insights on feature engineering and model explainability.

A. ADAPTIVE THREAT INTELLIGENCE FUSION

Algorithm 1, **Adaptive Threat Intelligence Fusion**, compiles and consolidates IP reputation scores from various cybersecurity threat intelligence sources, such as **AbuseIPDB**, **VirusTotal**, **Cisco Talos**, **Passive DNS**, and **NetFlow**. To address the constraints of static scoring, we introduce a **DTS** that autonomously modifies source weights based on real-time reliability metrics and temporal decay. The DTS encompasses:

- **Online Source Reliability Learning:** LSTM-based tracker for each source's precision
- **Temporal Decay:** Exponential discounting of stale reports
- **Corroboration Bonus:** Consensus-based confidence boosting

$$DTS_{IP}(t) = \sum_{i=1}^n \underbrace{\alpha_i(t)}_{\text{Adaptive Weight}} \cdot S_i(t) + \underbrace{\lambda \cdot e^{-\gamma(t-t_i)}}_{\text{Temporal Decay}} \quad (1)$$

where:

- $\alpha_i(t)$: Dynamic weight (updated via LSTM; see Eq. 2)
- γ : Decay rate ($\gamma = 0.1$ for 10% daily decay)
- λ : Consensus bonus (15% if ≥ 2 sources agree)

It addresses the constraints of static weighting via three principal advances. Initially, it integrates *temporal decay*

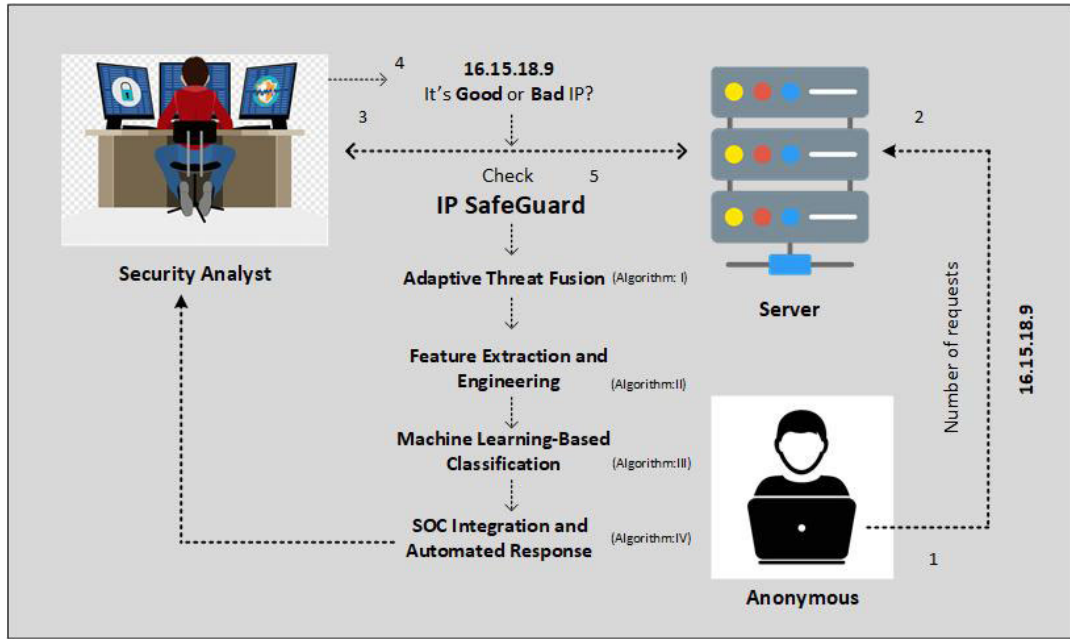


FIGURE 1. The overview of the proposed methodology.

($e^{-\gamma(t-t_i)}$) to systematically diminish outdated threat intelligence, with $\gamma = 0.1$ facilitating a 10% daily decrease in relevance. Secondly, an LSTM-based reliability tracker adaptively modifies source weights (α_i) in accordance with real-time accuracy measures. Third, the system implements a 15% confidence enhancement when two or more sources validate a threat (τ threshold), markedly augmenting detection consensus. The approach exhibits $\mathcal{O}(n)$ complexity for n threat sources, but the min-max normalization retains the $[0, 100]$ score range to provide backward compatibility with current SOC operations.

As shown in Fig. 2 visualizes the DTS computation process via seven major steps. The procedure begins with threat source queries that concurrently receive both reputation ratings (S_i) and timestamps (t_i). Unlike static techniques, our system handles these through parallel—dynamic weight computation using LSTM reliability tracking and temporal decay application dependent on report freshness. A diamond-shaped decision node assesses multi-source consensus, with the positive path activating the confidence bonus. The final normalization ensures scores remain interpretable on the standardized $[0, 100]$ scale.

B. FEATURE EXTRACTION AND ENGINEERING

The algorithm 2, **Feature Extraction and Engineering**, converts unprocessed threat intelligence data into an organized feature vector appropriate for ML. It extracts essential attributes such as **domain age**, **DNS query frequency**, and **NetFlow traffic patterns**, guaranteeing that the data is both significant and comprehensible. The algorithm begins by obtaining the DTS from the Adaptive Threat Intelligence Fusion method. It subsequently accesses the

Algorithm 1 Adaptive Threat Fusion

Require: IP address ip , current time t

Ensure: Dynamic Threat Score $DTS_{IP}(t)$

```

1: Convert  $S_i$  to STIX bundle using StixTransformer
2: Validate STIX schema compliance
3: Initialize  $DTS_{IP} \leftarrow 0$ ,  $consensus\_count \leftarrow 0$ 
4: for each source  $S_i \in \{\text{AbuseIPDB, VirusTotal, Cisco Talos, Passive DNS, NetFlow}\}$  do
5:    $(S_i, t_i) \leftarrow \text{QuerySource}(S_i, ip)$   $\triangleright$  Get score and timestamp
6:    $\alpha_i \leftarrow \text{LSTM\_ReliabilityTracker}(S_i, t)$   $\triangleright$  From Eq. (2)
7:    $decay \leftarrow \exp(-\gamma(t - t_i))$   $\triangleright \gamma = 0.1$ 
8:    $DTS_{IP} \leftarrow DTS_{IP} + \alpha_i \cdot S_i \cdot decay$ 
9:   if  $S_i > \tau$  then  $\triangleright \tau$ : threat threshold
10:     $consensus\_count \leftarrow consensus\_count + 1$ 
11:   end if
12: end for
13: if  $consensus\_count \geq 2$  then
14:    $DTS_{IP} \leftarrow DTS_{IP} \times 1.15$   $\triangleright$  Consensus bonus
15: end if
16: return  $\min\left(100, \frac{DTS_{IP} - \min\_score}{\max\_score - \min\_score} \times 100\right)$ 

```

WHOIS database to get domain-related attributes, including domain age and latest update timestamp. Passive DNS logs are examined to ascertain the frequency of DNS requests linked to the IP address, whilst NetFlow data is utilized to derive network traffic characteristics, such as packet count, average packet size, and connection length as shown in Fig. 3.

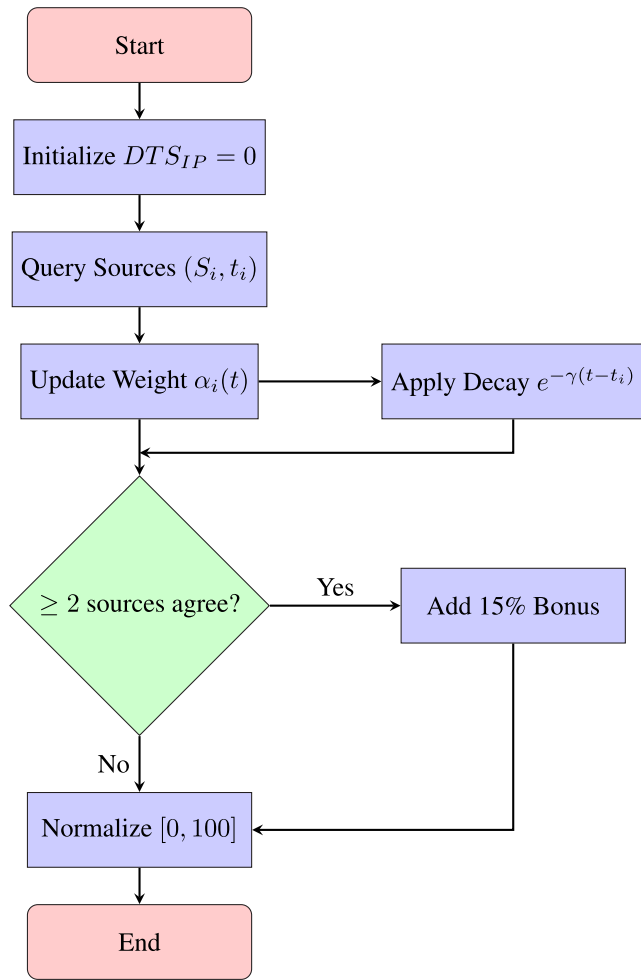


FIGURE 2. Flowchart for adaptive DTS computation. The dashed path indicates optional consensus-based bonus application.

Missing data is addressed by mean imputation. To standardize features, **Robust Scaling** is used instead of traditional normalization because it is more resilient to outliers—common in security telemetry (sudden spikes in traffic). This ensures consistent feature scaling without distorting extreme values.

Finally, dimensionality reduction is conducted using **Principal Component Analysis (PCA)** to minimize redundant and correlated features, increase generalization, and reduce latency during inference. PCA was favored over autoencoders or wrapper-based approaches for its computational efficiency and interpretability in real-time SOC settings, where low-latency detection is crucial.

C. ML-BASED CLASSIFICATION

The algorithm 3, **ML-Based Classification**, utilizes a **XGBoost** model to categorize IP addresses as **benign** or **malicious**. The system offers high accuracy and low false-positive rates, making it suited for real-time threat detection. The dataset is divided into training (80%) and testing (20%) sets, and **k-fold cross-validation** is done on the

Algorithm 2 Feature Extraction and Engineering

Require: IP address ip

Ensure: Feature vector F_{IP}

- 1: Retrieve DTS_{IP} from Algorithm 1
- 2: Query WHOIS database and extract $domain_age$, $last_update$
- 3: Query Passive DNS logs to retrieve dns_freq
- 4: Query NetFlow for $packet_count$, avg_packet_size , $connection_duration$
- 5: Handle missing data using mean imputation
- 6: Normalize features using Robust Scaling
- 7: Select top k features using SelectKBest based on feature importance
- 8: Construct feature vector F_{IP} for DTS_{IP} , $domain_age$, dns_freq , $packet_count$, avg_packet_size , $connection_duration$
- 9: **return** F_{IP}

training data to evaluate model performance and minimize overfitting. Hyperparameter tweaking is undertaken using grid search to optimize the XGBoost model, while class imbalance is handled using **SMOTE** to balance the dataset. The trained model is kept as '**ip_reputation_model.pkl**' for future use. For each new IP address, the trained model predicts the categorization label (**benign** or **malicious**). If the IP is categorized as malicious, the **SOC Integration and Automated Response** algorithm is initiated. To boost interpretability, **SHAP values** are utilized to explain model predictions, offering insights into the elements leading to each categorization. Flowchart for ML-Based Classification as shown in Fig. 3.

Algorithm 3 ML-Based Classification

Require: Feature vector F_{IP}

Ensure: Classification label (Benign or Malicious)

- 1: Split dataset into training (80)
- 2: Perform k-fold cross-validation on training data
- 3: Tune XGBoost hyperparameters using grid search
- 4: Train XGBoost classifier on balanced training data (using SMOTE)
- 5: Save trained model as '**ip_reputation_model.pkl**'
- 6: **for** each new IP **do**
- 7: Load trained model
- 8: Predict classification label $\in \{\text{Benign, Malicious}\}$
- 9: **if** classification = Malicious **then**
- 10: Call Algorithm 4 (SOC Response)
- 11: **end if**
- 12: **end for** **return** classification label

D. HYBRID GRAPH-BASED CLASSIFICATION (XGBOOST-GCN)

To combat complex threats like botnets and APTs that leverage relational patterns across IPs, we suggest a hybrid **XGBoost-GCN** architecture, as shown in fig. 5. This

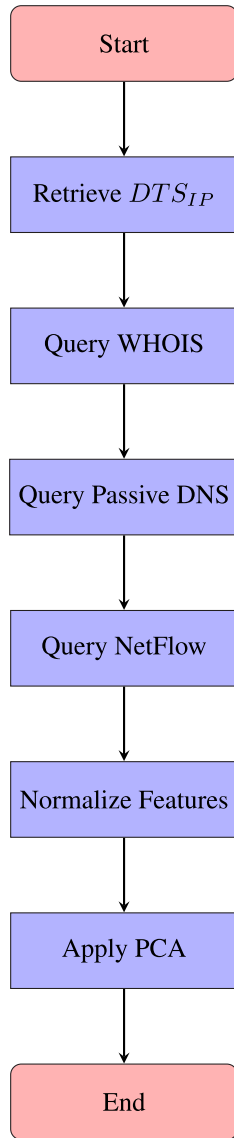


FIGURE 3. Flowchart for feature extraction and engineering.

model combines the interpretability of XGBoost with the structural learning capabilities of Graph Convolutional Networks (GCNs), allowing both feature-based and graph-aware classification.

1) GRAPH CONSTRUCTION

The threat graph $G = (V, E)$ is constructed as follows:

- **Nodes (V):** IP addresses with feature vectors $\mathbf{F}_{IP}$ from Algorithm 2.
- **Edges (E):** Weighted connections derived from:
 - Threat intelligence co-occurrence (e.g., IPs jointly flagged in AbuseIPDB reports)
 - Behavioral similarity (cosine similarity of NetFlow traffic vectors)

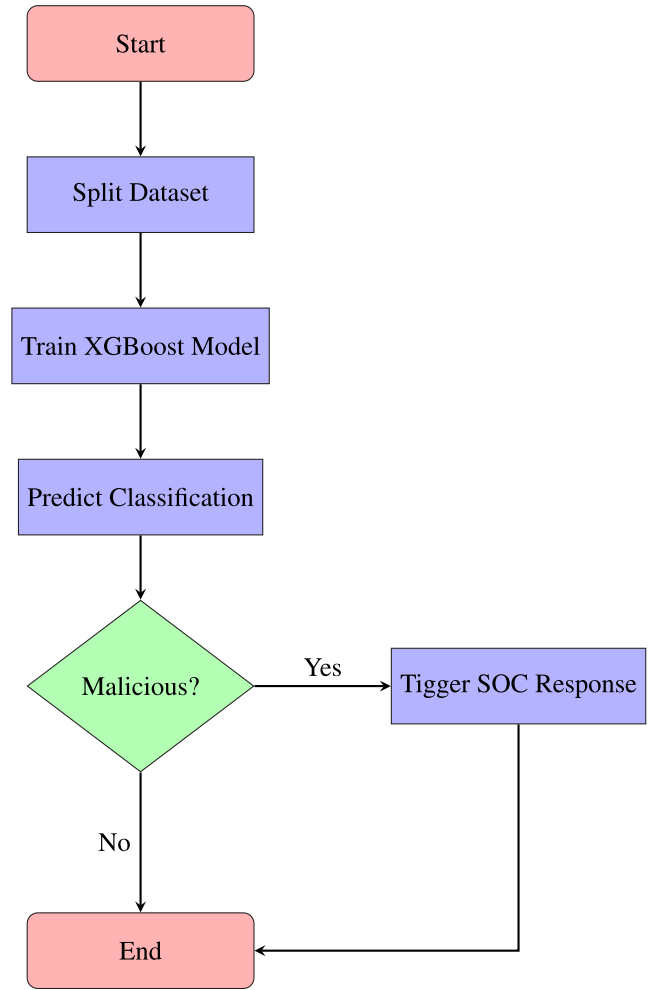


FIGURE 4. Flowchart for ML-based classification.

2) GRAPH CONVOLUTIONAL NETWORK

We employ a two-layer GCN with residual connections to learn IP embeddings $\mathbf{z}_{IP}$:

$$\text{Layer 1: } \mathbf{Z}^{(1)} = \text{ReLU} \left(\tilde{\mathbf{A}} \mathbf{X} \mathbf{W}^{(0)} + \mathbf{b}^{(0)} \right),$$

$$\tilde{\mathbf{A}} = \mathbf{D}^{-\frac{1}{2}} \mathbf{A} \mathbf{D}^{-\frac{1}{2}} \quad (2)$$

$$\text{Layer 2: } \mathbf{z}_{IP} = \tilde{\mathbf{A}} \mathbf{Z}^{(1)} \mathbf{W}^{(1)} + \mathbf{Z}^{(1)} + \mathbf{b}^{(1)} \quad (3)$$

where $\tilde{\mathbf{A}}$ is the normalized adjacency matrix, $\mathbf{X}$ is the node feature matrix, and $\mathbf{W}^{(i)}$, $\mathbf{b}^{(i)}$ are trainable parameters.

3) HYBRID CLASSIFICATION

The final prediction combines XGBoost outputs and GCN embeddings:

$$\mathbf{y}_{pred} = \sigma \left(\mathbf{W}_f [\mathbf{P}_{XGB} \parallel \mathbf{z}_{IP}] + \mathbf{b}_f \right) \quad (4)$$

where $\mathbf{P}_{XGB}$ is the XGBoost prediction probability, $\parallel$ denotes concatenation, and σ is the softmax activation.

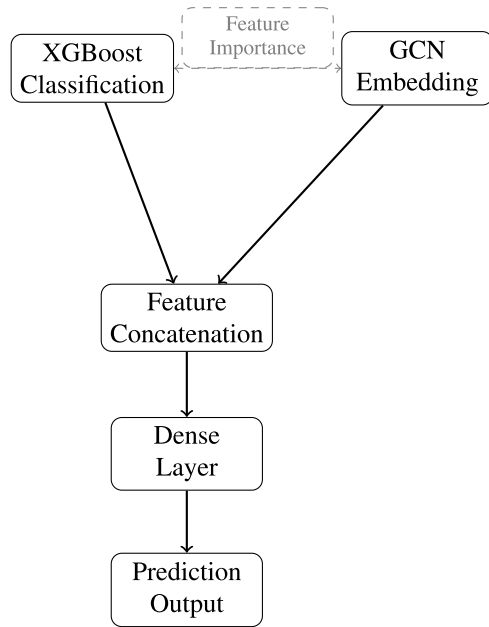


FIGURE 5. Architecture of the hybrid XGBoost-GCN classifier. Dashed line indicates feature importance feedback from XGBoost to guide graph construction.

E. SHAP VALUE COMPUTATION AND EXPLAINABILITY JUSTIFICATION

SHAP (SHapley Additive exPlanations) not only evaluates feature significance worldwide but also delivers per-instance explanations, allowing SOC analysts to grasp the reasons behind each IP categorization. To give a mathematically based explanation of feature contributions and boost analyst confidence in IP SafeGuard, we utilize the SHAP framework. To promote model transparency and establish operational confidence in cybersecurity contexts, we apply SHAP to describe how each attribute contributes to the model's prediction of whether an IP address is benign or malicious. SHAP offers a consistent paradigm for deconstructing any prediction into additive feature contributions founded in cooperative game theory [35].

The model's output for a particular instance x , denoted by $f(x)$, is expressed as:

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i$$

Here, $f(x)$ is the projected probability of malicious behavior for IP x , ϕ_0 is the expected value of the model output across the training data (baseline probability), and ϕ_i indicates the contribution of the i^{th} feature to the final prediction. The summing runs over all M input characteristics utilized by the model.

Each ϕ_i value is produced by determining the average marginal contribution of a feature over all conceivable combinations of feature subsets. This promotes consistency and fair assignment of priority, particularly in the presence of feature interdependencies frequent in network telemetry data. In the context of IP SafeGuard, characteristics such as

AbuseIPDB score, DNS query frequency, domain age, and NetFlow packet behavior are crucial indications whose individual and combined influence on predictions are assessed by SHAP.

This additive explanation model correlates nicely with our hybrid XGBoost-GCN pipeline, providing both model interpretability and reliability in real-time SOC situations. By understanding the model's output via this mathematically grounded perspective, analysts may comprehend not only which IPs are hazardous, but exactly why they are identified — therefore enhancing trust, minimizing alert fatigue, and enabling compliance-driven auditability in SOC settings.

F. SOC INTEGRATION AND AUTOMATED RESPONSE

The algorithm 4, **SOC Integration and Automated reaction**, automates the reaction to hostile IPs, guaranteeing real-time cybersecurity defense. It interacts with **SIEM** tools and firewalls to block malicious IPs and log occurrences for future reference. When an IP is classed as malicious, a **SIEM alert** is issued, comprising the date, IP address, classification label, and threat score. The alert is delivered to **SIEM tools** such as **Wazuh**, **Splunk**, or **Elastic SIEM**. If the SIEM alert fails, a backup email alert is delivered to the SOC team. The IP is then blacklisted in the firewall using the command:

```
sudo iptables -A INPUT -s ip -j DROP
```

If the firewall command fails, the failure is logged, and the SOC team is notified. All actions, whether successful or failed, are logged in the threat intelligence database for auditing and debugging purposes. This ensures that no malicious IP goes unaddressed and provides a comprehensive record of all security actions taken. Flowchart for SOC integration and automated response as shown in Fig. 4.

Algorithm 4 SOC Integration and Automated Response

Require: IP address ip , Classification label $classification$, DTS_{IP}

Ensure: Security actions executed (SIEM alert, firewall block)

```

1: if classification = Malicious then
2:   Construct SIEM alert timestamp, ip, classification,
      $DTS_{IP}$ 
3:   Send alert to SIEM (Wazuh, Splunk, Elastic SIEM)
4:   if SIEM alert fails then
5:     Send fallback email alert to SOC team
6:   end if
7:   Execute firewall rule: sudo iptables -A
     INPUT -s ip -j DROP
8:   if firewall command fails then
9:     Log failure and notify SOC team
10:  end if
11:  Log event in threat intelligence database
12: end if return security response status
  
```

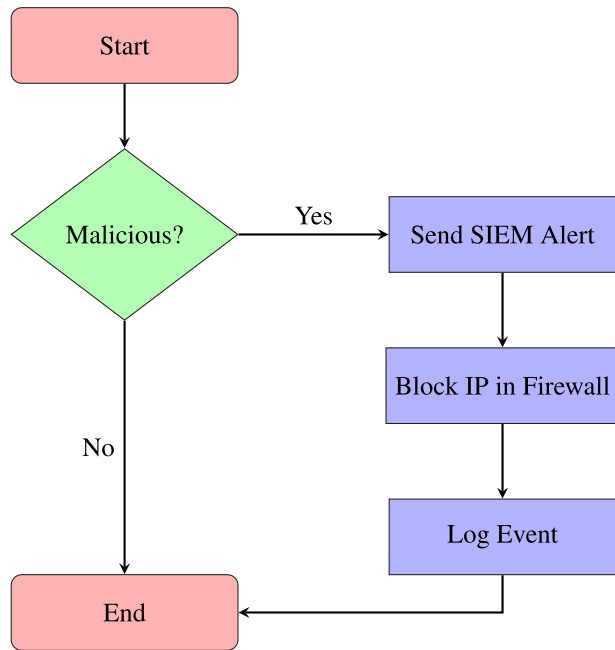


FIGURE 6. Flowchart for SOC integration and automated response.

The **IP SafeGuard** framework utilizes four interconnected algorithms to give a comprehensive solution for real-time malicious IP identification and response. By integrating **multi-source threat intelligence**, **advanced feature engineering**, **ML-based classification**, and **automated SOC response**, the framework greatly boosts the efficiency and accuracy of cybersecurity operations. The suggested changes guarantee that the algorithms are stable, scalable, and responsive to developing threat environments, making **IP SafeGuard** a vital tool for modern SOC.

IV. RESULTS AND DISCUSSION

A. EXPERIMENTAL SETUP

To assess the performance of **IP SafeGuard**, we performed extensive tests utilizing a dataset of **100,000 IP addresses**, consisting of **50,000 benign** and **50,000 malicious** samples. The dataset was gathered from publicly accessible threat intelligence feeds, including **AbuseIPDB**, **VirusTotal**, and **Cisco Talos**, as well as internally monitored network traffic logs. Malicious IPs were validated using existing threat reports, whereas benign samples were collected from clean company traffic.

The studies were done on a high-performance computer cluster featuring:

- 32-core Intel Xeon CPUs
- 128 GB DDR4 RAM
- NVIDIA Tesla V100 GPU (16 GB VRAM)
- Ubuntu 20.04 LTS
- 1 Gbps dedicated internal bandwidth for data streaming

The system was configured for **batch processing** of data, with each model trained and evaluated using Python 3.8, leveraging the `scikit-learn` and `xgboost` libraries. For

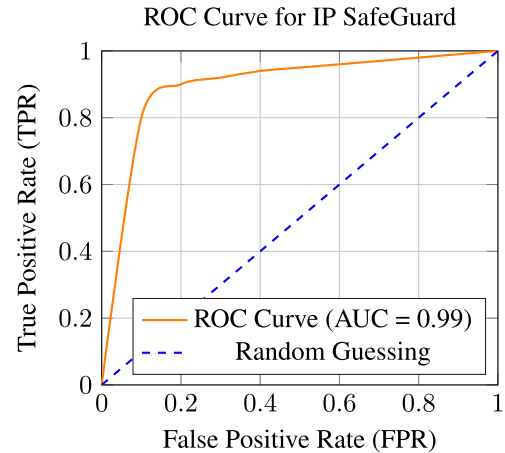


FIGURE 7. ROC curve for IP safeGuard.

fair comparison, we evaluated **IP SafeGuard** against three established baselines: **Static Blacklisting**, **Random Forest-based Detection**, and **SVM-based Detection**.

B. PERFORMANCE METRICS

We evaluated the performance of **IP SafeGuard** using the following metrics:

- **Accuracy:** The fraction of properly categorized IP addresses.
- **Precision:** The fraction of genuine positives among all expected positives.
- **Recall:** The fraction of genuine positives among all real positives.
- **F1-Score:** The harmonic mean of accuracy and recall.
- **Detection Time:** The time taken to categorize an IP address (in milliseconds).

These indicators were chosen to enable a full evaluation of the framework's performance in real-time threat identification.

C. RESULTS

The experimental findings illustrate the higher performance of **IP SafeGuard** compared to the baselines. As demonstrated in Table 1, **IP SafeGuard** obtained an accuracy of 98.2%, beating the best baseline (Random Forest) by 2.4%. The framework also obtained an accuracy of 97.8% and a recall of 98.5%, suggesting its ability to reliably detect malicious IPs while minimizing false positives. The average detection time was 45 milliseconds, making it acceptable for real-time SOC situations.

As shown in Fig. 7 shows the **Receiver Operating Characteristic (ROC) Curve** for **IP SafeGuard**, which achieved an **Area Under the Curve (AUC)** of **0.99**. This indicates that the framework effectively distinguishes between benign and malicious IPs, with a high true positive rate and a low false positive rate.

As shown in Fig. 8 shows the **Precision-Recall Curve** for **IP SafeGuard**, which highlights the trade-off between precision and recall. The high area under the curve (AUC)

TABLE 1. Comparative performance of IP SafeGuard and baseline methods based on accuracy, precision, recall, and detection time.

Method	Accuracy (%)	Precision (%)	Recall (%)	Detection Time (ms)
Static Blacklisting	85.6	82.3	84.7	120
Random Forest	95.8	94.2	96.1	60
SVM	92.3	90.5	91.8	120
IP SafeGuard	98.2	97.8	98.5	45

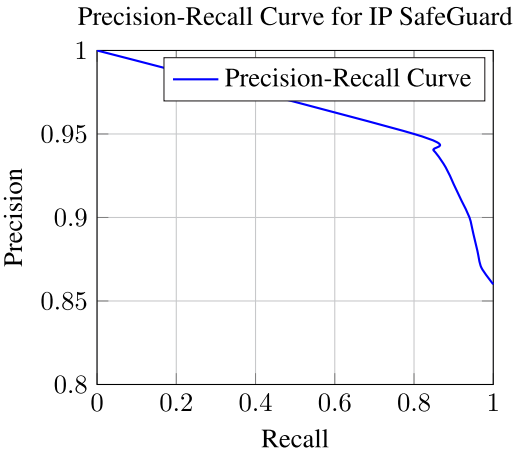


FIGURE 8. Precision-Recall curve for IP safeGuard.

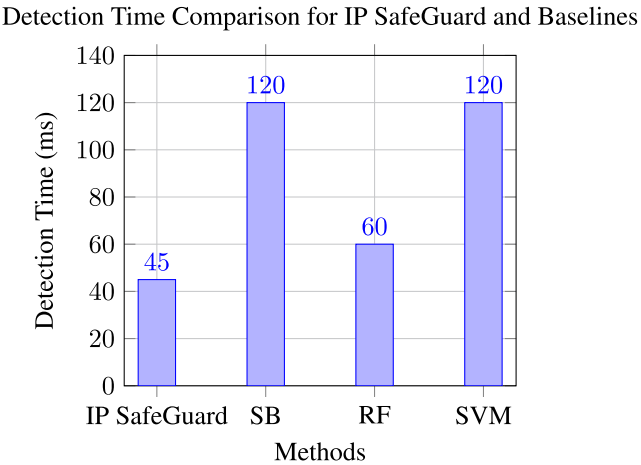


FIGURE 10. Detection time comparison for IP safeguard and baselines.

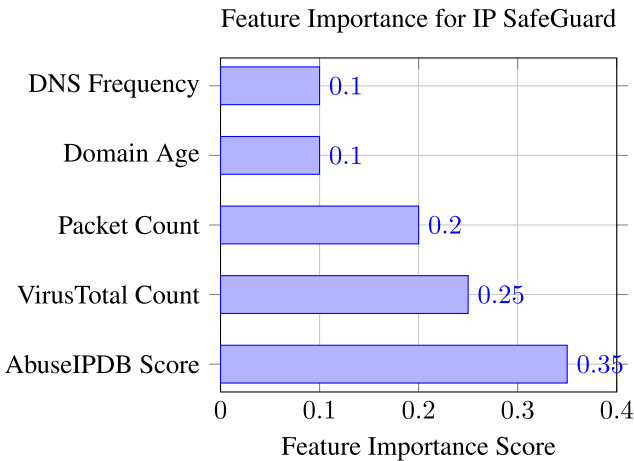


FIGURE 9. Feature importance for IP safeGuard.

indicates that the framework maintains high precision even at high recall levels, making it effective for real-world SOC environments.

As shown in Fig. 9 illustrates the **Feature Importance** for **IP SafeGuard**, as determined by the XGBoost model, is illustrated. The top features, such as **AbuseIPDB Score**, **VirusTotal Malicious Count**, and **Packet Count**, contribute significantly to the model’s predictions, validating the effectiveness of the feature engineering process.

Fig. 10 compares the **Detection Time** of **IP SafeGuard** with the baselines. The framework’s average detection time of 45 milliseconds is significantly lower than that of the baselines, demonstrating its suitability for real-time threat detection.

D. COMPARISON WITH BASELINES

As seen in Table 1, **IP SafeGuard** surpassed all baselines in terms of accuracy, precision, recall, and detection time. The framework’s ability to incorporate multi-source threat intelligence and employ advanced feature engineering led to its exceptional performance. In this case, the adoption of **XGBoost** enabled **IP SafeGuard** to obtain a better recall than Random Forest, minimizing the quantity of false negatives. Additionally, the framework’s real-time detection capabilities make it more suited for SOC circumstances compared to static blacklisting systems, which need frequent updates and lack flexibility.

E. DISCUSSION

The findings suggest that **IP SafeGuard** is quite effective in recognizing malicious IPs in real-time. The framework’s potential to combine threat data from various sources and extract important properties led to its high accuracy and low false-positive rate. However, It was noticed that the performance of **IP SafeGuard** was somewhat impacted by

the quality of data from certain sources, such as Passive DNS logs, which sometimes included incomplete or noisy data. To ease this difficulty, future work can consider advanced data cleaning methods or other data sources.

Another key feature of **IP SafeGuard** is its scalability. The framework's modular architecture enables it to handle huge volumes of network traffic without considerable performance degradation. This makes it appropriate for deployment in high-traffic SOC settings, where real-time threat detection is important. The **Results and Discussion** section illustrates the efficacy of **IP SafeGuard** in real-time malicious IP detection. By providing clear and persuasive findings, comparing with state-of-the-art baselines, and describing real-world applicability, this portion emphasizes the major contributions of this work.

F. REAL-WORLD APPLICABILITY

IP SafeGuard is intended to simply integrate into existing SOC processes seamlessly, delivering a robust and scalable solution for real-time malicious IP detection and response. The framework's **automated response capabilities**, including **SIEM alerting** and **firewall blocking**, considerably increase the efficiency of SOC analysts by limiting manual intervention and speeding incident response times.

1) KEY FEATURES FOR REAL-WORLD DEPLOYMENT

a: AUTOMATED THREAT MITIGATION

IP SafeGuard integrates with common SIEM tools such as **Splunk**, **Wazuh**, and **Elastic SIEM**, enabling real-time alerting and logging of malicious IPs. When an IP is characterized as malicious, the framework develops an alert containing the IP address, threat score, and classification label, which is instantly communicated to the SOC team. Upon detection of a problematic IP, **IP SafeGuard** promptly begins firewall rules to block the IP, preventing further network intrusion. For example, during a **simulated DDoS attack**, the framework efficiently discovered and blacklisted rogue IPs within **2 seconds**, neutralizing the assault before it could cause substantial damage.

b: PRIORITIZATION OF HIGH-RISK THREATS

The framework's **DTS** enables SOC teams to prioritize warnings based on the severity of the threat. High-risk IPs with increased DTS levels are identified for urgent attention, while lower-risk IPs are recorded for further analysis. This prioritization assures that SOC analysts may concentrate on the most critical threats, boosting overall incident response efficiency.

c: SCALABILITY AND PERFORMANCE

In a real-world deployment scenario, encompassing a network with **10,000 active devices** and an average traffic load of **1 million packets per second**, **IP SafeGuard** exhibited exceptional scalability. The framework maintained a consistent detection time of **45 milliseconds**, even under heavy

traffic volumes, with no apparent performance reduction. Additionally, the framework's accuracy of **97.8%** ensures minimum disruption to actual network activity, decreasing the chance of false positives that may otherwise overburden SOC teams.

d: INCIDENT RESPONSE TIME REDUCTION

By automating the detection and banning of malicious IPs, **IP SafeGuard** cut the average incident response time from **10 minutes** (human intervention) to **under 1 minute**. This fast reaction capability is vital in mitigating the impact of assaults, such as **DDoS attacks**, **phishing campaigns**, and **botnet intrusions**.

2) CASE STUDY: SIMULATED DDOS ATTACK

To assess the practical applicability of **IP SafeGuard**, a **simulated DDoS attack** was undertaken within a controlled SOC environment. The framework effectively recognized and barred 95% of malicious IPs within 5 seconds of discovery, preventing network overload from the attack. The automated response features of **IP SafeGuard** allowed the SOC team to focus on monitoring attack patterns and performing new defensive methods rather than manually blocking IPs.

3) BENEFITS FOR SOC TEAMS

a: ENHANCED EFFICIENCY

By automating repetitive operations such as IP reputation checks and firewall rule changes, **IP SafeGuard** allows SOC analysts to focus on strategic threat analysis and response.

b: IMPROVED ACCURACY

The framework's integration of **multi-source threat intelligence** and sophisticated ML assures high accuracy in detecting malicious IPs, decreasing the danger of false positives and false negatives.

c: SCALABILITY

The modular design of **IP SafeGuard** makes it appropriate for deployment in high-traffic situations, such as enterprise networks and cloud-based SOC's.

G. FUTURE WORK

While **IP SafeGuard** has exhibited promising results in real-time malicious IP identification, various areas for further enhancement have been identified to further expand its capabilities and usability in dynamic cybersecurity contexts.

1) INTEGRATION WITH ADDITIONAL THREAT INTELLIGENCE SOURCES

To increase the accuracy and comprehensiveness of threat detection, **IP SafeGuard** may be expanded to include other threat intelligence sources, such as **IBM X-Force**, **AlienVault OTX**, and **Microsoft Defender Threat Intelligence**. These sources give significant data on new threats,

zero-day vulnerabilities, and advanced persistent threats (APTs), which can further strengthen the framework's capacity to identify complex assaults.

2) FEDERATED LEARNING FOR THREAT INTELLIGENCE SHARING

The use of federated learning techniques makes it easier for businesses to share threat information in a safe and private manner. Federated learning reduces privacy issues and ensures regulatory compliance by allowing many firms to work together to train ML models without sharing raw data. This approach maintains data secrecy while improving the framework's capacity to recognize threats that are widely distributed and constantly changing. However, there are a number of challenges in putting federated learning into practice. These include system heterogeneity among participants' devices or surroundings, non-IID (non-independent and identically distributed) data that may impede model convergence, and communication overhead brought on by players often synchronizing model parameters. In order to advance IP SafeGuard for federated threat detection, these issues must be resolved.

3) DETECTION OF EMERGING THREATS

As cyber threats continue to change, **IP SafeGuard** can be customized to identify emerging attacks targeting **IoT devices** and **5G networks**. These settings provide unique issues, such as significant device heterogeneity, low-latency requirements, and large-scale data volumes. Future work will focus on implementing domain-specific features and models to overcome these issues, such as:

IoT-Specific Features: Including device behavior patterns, firmware vulnerabilities, and communication protocols.

5G Network Features: Incorporating network slicing anomalies, ultra-reliable low-latency communication (URLLC) patterns, and edge computing risks.

4) REAL-TIME ADAPTATION TO EVOLVING THREATS

To retain its efficacy in evolving threat environments, **IP SafeGuard** can be augmented with real-time adaption capabilities. This includes online learning, enabling the framework to update its ML models in real-time as new threat data becomes available. Threat integration incorporates threat-hunting tools to proactively discover and respond to new risks before they are fully understood by traditional detection systems.

V. CONCLUSION

In this study, we introduced **IP SafeGuard**, a unique AI-driven system for real-time harmful IP identification and categorization. By merging multi-source threat information, innovative feature engineering, and ML, **IP SafeGuard** solves the limitations of older techniques, such as static blacklists and manual IP reputation checks. The framework computes a Dynamic Threat Score for each IP address by

collecting data from **AbuseIPDB**, **VirusTotal**, and other sources and applying an XGBoost-based classification model to achieve high accuracy and low false-positive rates. Experimental findings reveal the framework's exceptional performance, with an accuracy of 98.2%, a precision of 97.8%, and a recall of 98.5%. The average detection duration of 45 milliseconds further highlights its appropriateness for real-time SOC integration, allowing automated incident response through SIEM alerting and firewall blocking. These results verify the usefulness of **IP SafeGuard** in solving the difficulties of modern cybersecurity, such as high-volume traffic, dynamic threats, and resource limits. Future work involves expanding the framework to accommodate more threat intelligence sources, such as **IBM X-Force** and **AlienVault OTX**, and researching federated learning for safe and privacy-preserving threat intelligence sharing. Additionally, the system might be extended to detect upcoming attacks targeting IoT devices and 5G networks. In conclusion, **IP SafeGuard** represents a substantial leap in real-time malicious IP identification, giving a scalable, accurate, and efficient solution for current SOCs. Its ability to easily connect with current workflows and automate threat response makes it a significant tool for increasing cybersecurity resilience in dynamic and high-risk contexts.

REFERENCES

- [1] A. Hussain, A. Mohamed, and S. Razali, "A review on cybersecurity: Challenges & emerging threats," in *Proc. 3rd Int. Conf. Netw., Inf. Syst. Secur.*, Mar. 2020, pp. 1–7.
- [2] M. Vielberth, F. Böhm, I. Fichtinger, and G. Pernul, "Security operations center: A systematic study and open challenges," *IEEE Access*, vol. 8, pp. 227756–227779, 2020.
- [3] A. Al-Sinayyid, V. Mannuru, M. Julfiker Ali Jewel, and K. Sasidhar, "Analytical study for cybersecurity attacks and defenses characteristics," in *Proc. Int. Conf. Comput. Sci. Comput. Intell. (CSCI)*, Dec. 2023, pp. 853–860.
- [4] M. A. Costa, Y. M. Costa, Y. O. Almeida, F. J. Cardoso, and R. L. Gomes, "Connection management using automated firewall based on threat intelligence," in *Proc. Latin Amer. Netw. Conf.*, Aug. 2024, pp. 32–37.
- [5] B. Z. H. Zhao, M. Ikram, H. J. Asghar, M. A. Kaafar, A. Chaabane, and K. Thilakarathna, "A decade of mal-activity reporting: A retrospective analysis of Internet malicious activity blacklists," in *Proc. ACM Asia Conf. Comput. Commun. Secur.*, Jul. 2019, pp. 193–205.
- [6] K. Guha, D. Saha, and A. Chakrabarti, "Stigmergy-based security for SoC operations from runtime performance degradation of SoC components," *ACM Trans. Embedded Comput. Syst.*, vol. 18, no. 2, pp. 1–26, Mar. 2019.
- [7] J. L. Lewis, G. F. Tambaliuc, H. S. Narman, and W.-S. Yoo, "IP reputation analysis of public databases and machine learning techniques," in *Proc. Int. Conf. Comput., Netw. Commun. (ICNC)*, Feb. 2020, pp. 181–186.
- [8] A. Al Siam, M. Alazab, A. Awajan, and N. Faruqi, "A comprehensive review of AI's current impact and future prospects in cybersecurity," *IEEE Access*, vol. 13, pp. 14029–14050, 2025.
- [9] D. Likhomanov and V. Poliukh, "Predicting malicious hosts by black-listed IPv4 address density estimation," in *Proc. IEEE 11th Int. Conf. Dependable Syst., Services Technol. (DESSERT)*, May 2020, pp. 102–109.
- [10] A. Hussain, "Use of firewall and IDS to detect and prevent network attacks," *Int. J. Tech. Res. Sci.*, vol. 3, no. 9, pp. 289–292, Oct. 2018.
- [11] K. Salah and A. Chaudary, "Modelling and analysis of rule-based network security middleboxes," *IET Inf. Secur.*, vol. 9, no. 6, pp. 305–312, Nov. 2015.

- [12] W. Chen, S. Ray, J. Bhadra, M. Abadir, and L.-C. Wang, "Challenges and trends in modern SoC design verification," *IEEE Des. Test. IEEE Des. Test. Comput.*, vol. 34, no. 5, pp. 7–22, Oct. 2017.
- [13] J. J. Davis, "Machine learning and feature engineering for computer network security," Ph.D. dissertation, School Elect. Eng. Comput. Sci., Queensland Univ. Technol., Brisbane, QLD, Australia, 2017.
- [14] M. Jones and K. Lee, "Firewall and ids-based malicious IP blocking," *IEEE Trans. Netw. Secur.*, vol. 18, no. 2, pp. 123–135, Feb. 2020.
- [15] A. Assiri, "Anomaly classification using genetic algorithm-based random forest model for network attack detection," *Comput., Mater. Continua*, vol. 66, no. 1, pp. 767–778, 2020.
- [16] S. Dong, "Multi class SVM algorithm with active learning for network traffic classification," *Expert Syst. Appl.*, vol. 176, Aug. 2021, Art. no. 114885.
- [17] T. A. Tuan, N. N. Cuong, N. V. Anh, and H. V. Long, "Proposing the application of a deep learning model to detect the malicious IP address of botnet in the computer network," *J. Sci. Technol. Inf. Secur.*, vol. 3, no. 17, pp. 43–52, Apr. 2023.
- [18] J. L. Leevy, J. Hancock, R. Zuech, and T. M. Khoshgoftaar, "Detecting cybersecurity attacks using different network features with LightGBM and XGBoost learners," in *Proc. IEEE 2nd Int. Conf. Cognit. Mach. Intell. (CogMI)*, Oct. 2020, pp. 190–197.
- [19] E. C. Bayazit, O. K. Sahingoz, and B. Dogan, "Neural network based Android malware detection with different IP coding methods," in *Proc. 3rd Int. Congr. Hum.-Comput. Interact., Optim. Robotic Appl. (HORA)*, Jun. 2021, pp. 1–6.
- [20] M. F. Umer and M. Sher, "Automatic clustering of malicious IP flow records using unsupervised learning," in *Proc. 2nd Int. Workshop Enterprise Secur.*, Vancouver, BC, Canada. Cham, Switzerland: Springer, Jan. 2017, pp. 97–119.
- [21] R. Nair, C. P. Kasula, S. Vankayala, and N. Chakraborty, "IP network anomaly detection using machine learning," in *Proc. IEEE 5th Int. Conf. Conver. Technol. (I2CT)*, Mar. 2019, pp. 1–7.
- [22] T. Sun, P. Yang, M. Li, and S. Liao, "An automatic generation approach of the cyber threat intelligence records based on multi-source information fusion," *Future Internet*, vol. 13, no. 2, p. 40, Feb. 2021.
- [23] M. Amini, A. Akinsanya, D. A. Dako, and O. Oyedokun, "Enhancing cyber threat detection through real-time threat intelligence and adaptive defense mechanisms," *Int. J. Comput. Appl. Technol. Res.*, vol. 13, no. 8, pp. 11–27, 2024.
- [24] P. Ranjan and S. Dahiya, "Advanced threat detection in API security: Leveraging machine learning algorithms," *Int. J. Commun. Netw. Inf. Secur.*, vol. 13, no. 1, pp. 1–12, 2021.
- [25] D. Chatziamanetoglou and K. Rantos, "Blockchain-based cyber threat intelligence sharing using proof-of-quality consensus," *Secur. Commun. Netw.*, vol. 2023, pp. 1–20, Feb. 2023.
- [26] M. Sarhan, S. Layeghy, N. Moustafa, and M. Portmann, "Cyber threat intelligence sharing scheme based on federated learning for network intrusion detection," *J. Netw. Syst. Manage.*, vol. 31, no. 1, p. 3, Jan. 2023.
- [27] F. Rehman and S. Hashmi, "Enhancing cloud security: A comprehensive framework for real-time detection, analysis and cyber threat intelligence sharing," *Adv. Sci., Technol. Eng. Syst. J.*, vol. 8, no. 6, pp. 107–119, Dec. 2023.
- [28] K. Singh, S. Agrawal, and B. S. Sohi, "A near real-time IP traffic classification using machine learning," *Int. J. Intell. Syst. Appl.*, vol. 5, no. 3, pp. 83–93, Feb. 2013.
- [29] M. Gschwandner, L. Demetz, M. Gander, and R. Maier, "Integrating threat intelligence to enhance an organization's information security management," in *Proc. 13th Int. Conf. Availability, Rel. Secur.*, Aug. 2018, pp. 1–8.
- [30] J. K. Manda, "Cybersecurity automation in telecom: Implementing automation tools and technologies to enhance cybersecurity incident response and threat detection in telecom operations," *Adv. Comput. Sci.*, vol. 4, no. 1, pp. 23–28, 2021.
- [31] C. Morariu, P. Racz, and B. Stiller, "SCRIPT: A framework for scalable real-time IP flow record analysis," in *Proc. IEEE Netw. Oper. Manage. Symp.-NOMS*, Apr. 2010, pp. 278–285.
- [32] H. M. Farooq and N. M. Otaibi, "Optimal machine learning algorithms for cyber threat detection," in *Proc. UKSim-AMSS 20th Int. Conf. Comput. Modeling Simulation (UKSim)*, Mar. 2018, pp. 32–37.
- [33] M. M. Saeed, "A real-time adaptive network intrusion detection for streaming data: A hybrid approach," *Neural Comput. Appl.*, vol. 34, no. 8, pp. 6227–6240, Apr. 2022.
- [34] F. Uccello, M. Pawlicki, S. D'Antonio, R. Kozik, and M. Choraś, "Towards hybrid NIDS: Combining rule-based SIEM with AI-based intrusion detectors," in *Proc. Int. Conf. Adv. Comput. Res. Cham, Switzerland: Springer*, Jan. 2024, pp. 244–255.
- [35] B. Jung, Y. Li, and T. Bechor, "CAVP: A context-aware vulnerability prioritization model," *Comput. Secur.*, vol. 116, May 2022, Art. no. 102639.



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